

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0740

Alkali 100 kV
July 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0740. The Interconnection Customer proposes a 143.2 MW solar photovoltaic facility interconnecting at the Alkali 100 kV substation in the Ohio area. The request was received on February 04, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 290 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,470,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ALKALI – URBANA 100 kV	WOLFE SUBSTATION – BARRETT 500 kV Line	132.7	140	55.6
APPLE VALLEY – SITHE 100 kV Line	WOLFE – BARRETT 500 kV	131.1	143	5.8
CARLEY RD – CARNAL 100 kV L	WOLFE – BARRETT 500 kV	122.1	146	7.9
IMPERIAL – CHAMBERLIN 230 kV	RAVENNA – HYLEBOS 500 kV	105.8	704	5.4
MEDORA – SIMPSON STUART TAP 345 kV Line	HAMPTON RD – MEDOA 345 kV	104.7	1,206	5.2
CALDWELL – BLACKROCK 345 kV	CARBON CENTER 500 kV/345 kV Transformer Bank	101.0	1,217	5.1
CARLEY RD – URBANA 100 kV L	WOLFE – BARRETT SUBSTATION 500 kV	100.5	107	34.3
PINEY CRK PROJECT 500-345 kV Transformer Bank	PINEY CRK PROJECT – HILTOP 500 kV	100.0	1,923	8.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ALKALI – URBANA 100 kV	Rebuild line with higher-capacity conductor	\$3,846,000	100% (\$3,846,000)
SITHE – APPLE VLY SUBSTATION 100 kV Line	Rebuild line with higher-capacity conductor	\$3,014,000	68% (\$2,062,000)
CARLEY RD – CARNL 100 kV	Rebuild line with higher-capacity conductor	\$21,744,000	100% (\$21,744,000)
IMPERIAL – CHAMBERLIN 230 kV	Reconductor line	\$5,378,000	1% (\$55,000)
SIMPSON STUARTTAP – MEDORA 345 kV	Reconductor line	\$11,308,000	2% (\$277,000)
CALDWELL – BLACKROCK 345 kV	Reconductor line	\$25,412,000	2% (\$438,000)
URBANA – CARLEY ROAD 100 kV	Reconductor line	\$843,000	100% (\$843,000)
PINEY CRK PROECT 500 kV/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$18,404,000	1% (\$205,000)
	Total allocated to project		\$29,470,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.